AquaSentinel: Raspberry Pi prototype engineering design

Version 1.0, 28 October 2026. Prepared by Team Sanjivani Hero's Journey.
Inputs: the team's three handwritten notes (main problems, component list, component purposes). They are treated as the definitive component selection. **Status:** a design to build and test. Nothing in it has been wired or measured yet.

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0. Summary

Verdict: the selected hardware can make a working AquaSentinel bench prototype for the finale (19–20 Nov 2026). It needs about **₹3,000 of small additions**: an analog-to-digital converter (ADC), voltage regulators, a level shifter, a real-time clock and switching parts. It can also run a short supervised field pilot. It is not yet a long-unattended field station: the main limits are power, 2G-only communication and sensor drift.

The ten decisions that matter most

1. **Add 2 × ADS1115 ADCs.** The Raspberry Pi has no analog inputs. pH, TDS, turbidity, MQ-3, the light sensor, the rain sensor and the battery voltage are all analog signals. [1][3]
2. **Keep every signal into the Pi and the ADC at or below 3.3 V.** The Pi's pins are not 5 V tolerant. The turbidity, MQ-3 and (likely) pH boards can output up to 4.5–5 V, so each gets a 12 kΩ / 20 kΩ divider and an RC filter. [1][3][7]
3. **Give the SIM800L its own 4.0 V supply rated for 2 A bursts, with a 1000 µF low-ESR capacitor.** Never power it from the Pi. Level-shift the Pi's transmit line, because the SIM800L accepts at most 3.1 V on its serial input. [4]
4. **Never power the pH and TDS boards at the same time.** Two powered probes in the same water disturb each other. Switch each board's supply and read them one after the other; isolator boards are the more expensive alternative. [17]
5. **Keep the "mini L298N" only for a pump of 6 V or less.** It is really an MX1508 chip (motor supply 2–10 V, 1.5 A). For a 12 V pump, use a logic-level MOSFET; the pump turns one way only, so it doesn't need an H-bridge. [14][13]
6. **Use the MQ-3 only as a vapour indicator in a sealed headspace above a sample,** as the team planned. It needs a 48 h first burn-in, its heater draws up to 0.9 W, and it reacts to alcohol and some solvents. It does not measure the water. [5]
7. **Each cycle reads all sensors first, then transmits and runs the pump.** That keeps GSM bursts and pump noise away from the analog readings.
8. **Protect the SD card and the clock.** Run a read-only root file system, the hardware watchdog and a DS3231 real-time clock, so power cuts don't corrupt the card or lose the time.
9. **Field power:**
    - With the Pi always on, the system needs about **100 Wh/day**: a 50 W panel and a 20 Ah battery.
    - Waking the Pi only every 15 minutes cuts that to about **19 Wh/day**, which fits the R3's 20 W panel and 6 Ah battery.
10. **Claims language:** call it a prototype built from available parts. The R3 field design (ESP32, industrial RS485 probes, LTE) stays the product the jury saw.

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1. Scope, inputs and assumptions

Problems the team wants to address (note 1):

- industrial discharge;
- agricultural runoff;
- sewage and wastewater;
- oil spills and leaks;
- plastic and solid waste.

Hardware the team has (notes 2 and 3):

#	Item (team's words)	Team's purpose
1	Raspberry Pi 3 Model B+ 1 GB	Main on-board computer
2	Rain sensor module (plate + module, "raindrop sensor")	Rain detection
3	Mini DC submersible pump, 3–12 V	Collect a 250 mL water sample
4	Mini L298N	Control the pump
5	LDR	Day/night tracking
6	MQ-3	VOCs, solvents or ethanol in the sealed air space above a contaminated sample
7	TDS Meter V1.0	Total dissolved solids
8	DHT11	Air temperature and humidity above the water
9	SIM800L GSM/GPRS ("red PCB module")	Send data, messages and alerts
10	DS18B20 digital temperature probe	Water temperature
11	pH sensor	Acidity / alkalinity, 0–14
12	Turbidity sensor	Water clarity (infrared scattering)

Assumptions (confirm them; section 15 lists what to check):

#	Assumption	Why it matters
A1	The pH sensor is the common **PH-4502C board with an E-201 BNC probe**, powered at 5 V. If it is DFRobot SEN0161-V2 (output 0–3.0 V), its divider can be removed.	Output range and calibration method
A2	"TDS Meter V1.0" is the **DFRobot SEN0244 board or a clone** (3.3–5.5 V in, 0–2.3 V out, 0–1000 ppm). [6]	Range and ADC channel
A3	The turbidity sensor is **DFRobot SEN0189 or a clone** (5 V, 0–4.5 V out, top not waterproof). [7]	Divider and housing
A4	The pump is the common **3–6 V mini submersible pump** (130–220 mA, 80–120 L/h). The note says 3–12 V; a 12 V pump changes the driver. [16]	Driver choice
A5	The "red PCB" SIM800L is the **bare mini module** (VBAT 3.4–4.4 V, no on-board regulator). A blue "SIM800L V2 5V" board has its own regulator and level shifting. [4]	Power and level shifting
A6	The rain and LDR modules are **LM393 comparator boards** with digital (DO) and analog (AO) outputs, working at 3.3–5 V. [15]	Can connect directly at 3.3 V
A7	Finale demo: indoors on mains power. Field pilot: a drain, canal or river bank, 1–2 weeks, supervised.	Power and enclosure sizing

2. What the system can and cannot detect

The sensors measure **physical and chemical indicators**. They do not identify specific pollutants. A change in indicators shows that something happened; the 250 mL sample tells the lab what it was.

Problem	Measured directly	Can only be inferred	Cannot detect with this hardware	Scientifically justified addition
Industrial discharge	TDS rise (dissolved salts and metal ions), pH out of range, water temperature rise (thermal discharge), turbidity	A discharge event, from a sudden multi-parameter change	Which chemical; heavy metals; COD	Lab analysis of the auto-collected sample. An EC sensor with a wider range than TDS (0–20 mS/cm)
Agricultural runoff	Turbidity rise (sediment), TDS rise	Runoff when the rain sensor says "raining" and turbidity or TDS jumps	Nitrate, phosphate, pesticides	Nitrate ion-selective electrode or lab tests
Sewage and wastewater	Turbidity, TDS, pH and temperature changes	Organic pollution, only weakly	Dissolved oxygen, ammonia, BOD, coliform bacteria	**Dissolved oxygen sensor:** the single most useful addition for sewage. It is already in the R3 design
Oil spills and leaks	Nothing directly. The MQ-3 responds weakly to some petrol vapours	A volatile fuel or solvent spill, from the headspace test	Heavy oils (diesel, crude); oil film thickness	Oil-in-water fluorescence sensor (expensive), or a camera watching for sheen
Plastic and solid waste	Nothing	Nothing	Floating plastic, debris	**Raspberry Pi Camera Module 3** with image capture, plus later object detection. The Pi is well suited to this

MQ-3 in particular:

- **What it detects:** alcohol vapour, 25–500 ppm in air, and it is cross-sensitive to other gases. [5]
- **The method that works:** fill a sample, seal the air space above it, let vapours equilibrate, then read the MQ-3. A rise means "volatile organics present". A distillery effluent, solvent or petrol spill is the plausible cause.
- **How to state it:** treat the reading as an indicator. Never report it as a concentration in water.

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3. Component analysis

3.1 Raspberry Pi 3 Model B+ (main computer)

- **Key specs:**
  - power supply 5 V, 2.5 A recommended; about 500 mA typical bare-board current;
  - GPIO at 3.3 V and **not 5 V tolerant**, up to 16 mA per pin and 50 mA for all pins together. [1]
- **No analog inputs**, so it needs an external ADC.
- **Interfaces used:**
  - I2C1 (GPIO2/3) for the two ADCs and the RTC;
  - 1-Wire (GPIO4) for the DS18B20;
  - a kernel driver (dtoverlay) for the DHT11;
  - the PL011 UART on GPIO14/15 for the SIM800L;
  - WiFi for MQTT;
  - GPIO for the pump, power switches, LED and button. [1][2][12]
- **Limitations:** about 2.5 W or more continuously; SD-card corruption on sudden power loss; no real-time clock. It needs shade in an enclosure.
- **Verdict:** a good choice for the prototype (Python, logging, dashboard, camera later). For a solar field unit, wake it on a timer (section 5.3).

3.2 ADS1115 ADC (to add)

- **Key specs:**
  - 16-bit, 4 single-ended inputs, I2C (addresses 0x48–0x4B), supply 2.0–5.5 V;
  - programmable gain, full-scale range ±0.256 V to ±6.144 V;
  - inputs must stay between ground and the supply voltage. [3]
- **Design:**
  - Power both ADCs from the Pi's 3.3 V pin, so the I2C lines stay at 3.3 V and need no level shifting.
  - Use the ±4.096 V range (125 µV per step) and keep every input below 3.3 V with dividers.
  - #1 at 0x48 reads the water sensors; #2 at 0x49 reads the housekeeping signals.

3.3 pH sensor (PH-4502C assumed)

- **Principle:** a glass electrode produces about 59 mV per pH unit at 25 °C. The board amplifies and offsets this so the output is near 2.5 V at pH 7 after the offset is trimmed. [9]
- **Supply:** 5 V.
- **Output:** analog Po. Listings disagree on the output range, so treat it as **up to 5 V** and divide it.
- **Calibration:**
  1. Offset: short the BNC input and trim the potentiometer until Po reads 2.50 V. [9]
  2. Two or three points in buffers pH 4.01, 6.86/7.00 and 9.18.
- **Limitations:**
  - The probe must be stored wet in KCl.
  - The glass bulb is fragile.
  - Readings drift, so calibrate before each demo day and weekly in the field.
  - Readings are temperature dependent; the DS18B20 corrects for this.
  - Other powered probes in the same water disturb the reading (section 6.3).

3.4 TDS Meter V1.0 (DFRobot SEN0244 type)

- **Key specs:** 3.3–5.5 V, 0–2.3 V output, 3–6 mA, **0–1000 ppm**, ±10 % of full scale at 25 °C, 2-needle waterproof probe. [6]
- **Conversion** (DFRobot sample code):
  - Temperature compensation: $k = 1 + 0.02 \times (T - 25\ °C)$, then $V' = V / k$.
  - TDS (ppm) = $(133.42\ V'^3 - 255.86\ V'^2 + 857.39\ V') \times 0.5$.
- **Limitations:**
  - TDS is estimated from conductivity, so it cannot tell which ions are present.
  - **The 1000 ppm ceiling is about 2,000 µS/cm**, below the R3's 2,250 µS/cm alarm level, so the TDS alarm must sit lower (section 7.4).
  - It disturbs the pH probe if both are powered at once.
- **Connection:** direct to the ADC (its output stays below 3.3 V), with an RC filter.

3.5 Turbidity sensor (SEN0189 type)

- **Principle:** infrared light is scattered by suspended particles; the output voltage **falls** as turbidity rises. [7]
- **Key specs:** 5 V, about 40 mA, analog 0–4.5 V. **The top of the probe is not waterproof.** [7]
- **Conversion:** DFRobot's approximate curve, valid for 2.5–4.2 V on a 5 V supply: NTU ≈ $-1120.4\ V^2 + 5742.3\ V - 4352.9$. It is not a certified NTU; use it for trends.
- **Limitations:**

    - Ambient light and bubbles affect the reading, so measure inside the housing, out of direct sun.
    - The optics foul and need cleaning.
    - Calibration against standards is only approximate.
- **Connection:** 12 k / 20 k divider (4.5 V → 2.8 V), then RC filter, then ADC.

3.6 DS18B20 waterproof probe

- **Key specs:** 3.0–5.5 V; −55 to +125 °C; ±0.5 °C from −10 to +85 °C; 9–12 bit resolution; 750 ms per 12-bit conversion; 1-Wire bus with a 4.7 kΩ pull-up. [10]
- **On the Pi:** the `w1-gpio` overlay on GPIO4; the readings appear under `/sys/bus/w1/devices/28-*`. [12]
- **Use:** water temperature, temperature compensation for pH and TDS, and thermal-discharge detection.

3.7 DHT11 (air temperature and humidity)

- **Key specs:** humidity 20–90 % ±5 %; temperature 0–50 °C ±2 °C; 3.3–5.5 V; a single-wire protocol with tight timing; read no faster than every 1–2 s. [11]
- **On the Pi:** use the kernel driver `dtoverlay=dht11,gpiopin=17`. Python bit-banging of this protocol is unreliable under Linux. [12]
- **Limitations:**
    - Close to water the humidity often exceeds 90 %, outside the DHT11's range.
    - Mount it in a louvered shade 0.5–1 m above the water.
- **Use:** context and correction for the MQ-3, plus the enclosure climate.

3.8 MQ-3 alcohol / VOC gas sensor

- **Key specs** (Winsen datasheet): [5]
    - heater 5.0 V ±0.1 V, about 29 Ω, **up to 900 mW**;
    - detection range **25–500 ppm alcohol**;
    - **preheat over 48 hours** before first use;
    - readings depend on temperature and humidity.
- **Module:** the sensor and a load resistor RL form a divider; the analog output reaches up to 5 V and needs dividing. The digital output (an LM393 threshold) is 5 V logic and is not used here.
- **Use:**
    1. Calculate Rs = RL × (5 V − Vout) / Vout.
    2. Find R0 in clean air after burn-in.
    3. Alarm on Rs/R0 dropping in the sealed headspace test.
    4. Measure RL on your module with a multimeter.
- **Limitations:** cross-sensitive to other vapours; slow recovery; high power. It is an indicator only.

3.9 Rain sensor (plate + LM393 module)

- **Function:** a nickel-plated comb, wet or dry. The module gives a digital and an analog output. [15]
- **Supply:** 3.3 V, so the digital output can go straight to a GPIO pin.
- **Limitations:** the comb **corrodes when powered continuously**. Power it only while reading, through a transistor switch.
- **Use:** context. A turbidity or TDS jump while it rains is flagged "runoff-likely" (agricultural runoff).

3.10 LDR (light-dependent resistor)

- **Function:** a photoresistor on an LM393 module.

- **Use:** the digital output gives day/night; the analog output gives a rough light level on ADC #2.
- **Limitations:** it is shaded by the panel and fooled by street lights. The RTC time plus a sunrise table is a more reliable day/night signal; keep the LDR as a sanity check and for dimming the status light at night.

3.11 Mini submersible pump

- **Typical** 3–6 V model: 130–220 mA, 80–120 L/h, lift 40–110 cm (some listings print mm by mistake), about 500 h rated life. **Must not run dry.** [16]
- **Fill time:** 250 mL at 100 L/h takes about 9 s. Calibrate it on the bench, because tubing and lift reduce the flow.
- **Placement:** the pump must sit in the water (with the intake strainer) and push up to the bottle; a 0.2 m lift is well within its range.

3.12 "Mini L298N" motor driver

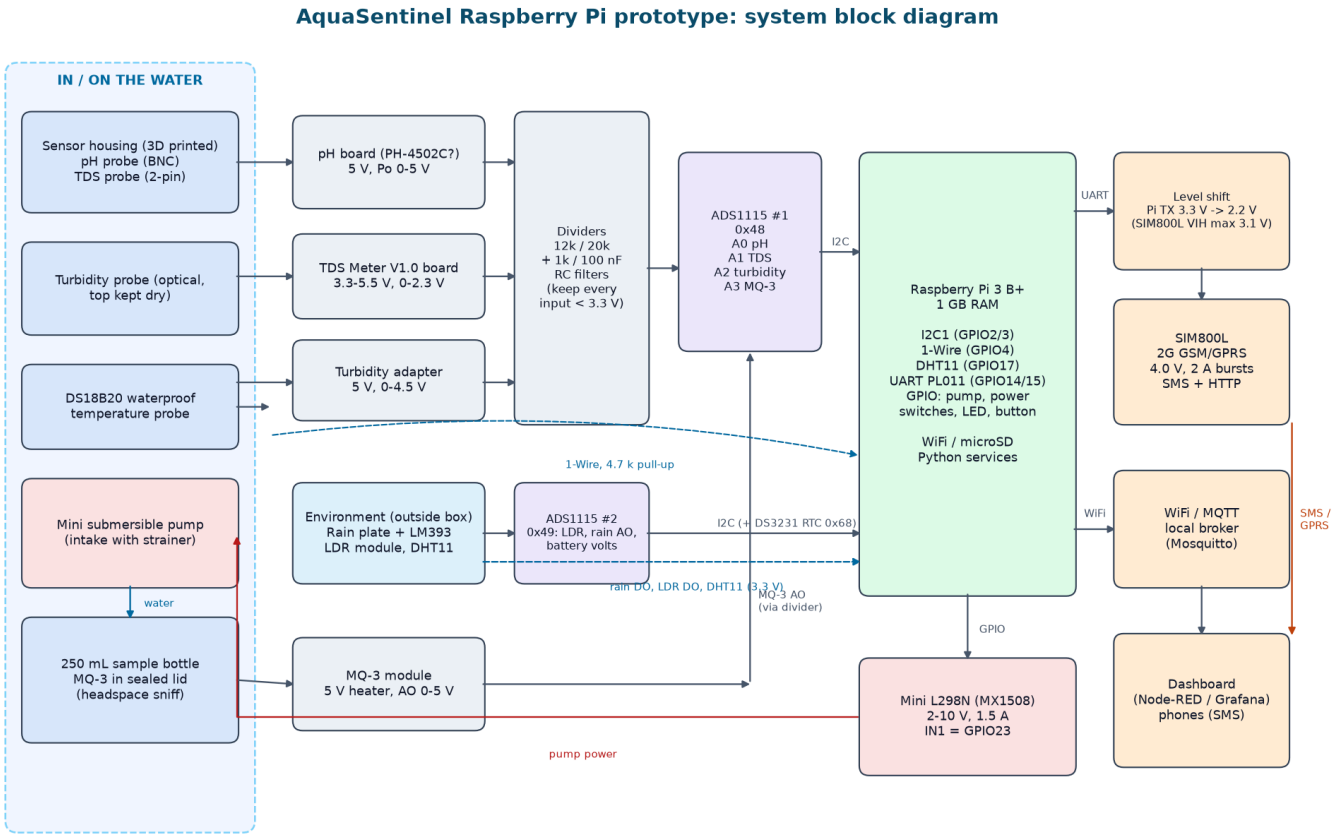
- **What it is:** these boards carry an **MX1508** chip (sometimes a clone):
  - motor supply 2–10 V, 1.5 A continuous;
  - logic inputs 1.8–7 V, so the Pi's 3.3 V GPIO is fine. [14]
- **Verdict:** suitable for a pump of 6 V or less. **Not suitable for a 12 V pump.**
- **Why not the full-size L298N:** it is a bipolar bridge rated to 46 V and 2 A per channel, but it loses roughly 2–3 V across the bridge. [13]
- **For one direction only,** the simplest option is a logic-level MOSFET (IRLZ44N or AO3400) with a flyback diode.

3.13 SIM800L GSM/GPRS

- **Key specs** (SIMCom Hardware Design V2.02): [4]
  - quad-band 2G GSM/GPRS;
  - **VBAT 3.4–4.4 V, 4.0 V recommended, up to 2 A bursts**; a low-ESR bypass capacitor (100 µF or more) is strongly recommended;
  - 17 mA idle; 175–530 mA average during GPRS transfer.
- **Serial levels:** input high 2.1–3.1 V, output high 2.8 V. **The Pi's 3.3 V transmit line exceeds the 3.1 V maximum**, so it needs a divider or level shifter. The SIM800L's own transmit line (2.8 V) is read correctly by the Pi.
- **Reset:** a 3.3 V GPIO can drive the RESET pin directly. [4]
- **Network:**
  - It is 2G only. In India, Jio has no 2G network. Vodafone Idea retired 3G but kept 2G, and Airtel still runs 2G; no shutdown announcement was found as of this research. [19]
  - Use an **Airtel or Vi SIM**, and check the signal strength at the site with `AT+CSQ`.
  - Upgrade path: the SIMCom A7670C LTE Cat-1 module from the R3 design.

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4. System architecture



Interface decisions

Concern	Decision
Analog inputs	2 × ADS1115 on I2C, powered at 3.3 V. Every input is limited to 3.3 V
Voltage levels	3.3 V everywhere at the Pi. 5 V module outputs are divided. The Pi's transmit line to the SIM800L goes through a 1 kΩ / 2 kΩ divider (3.3 → 2.2 V)
pH and TDS in the same water	Switch each board's supply and never power both at once. Read them in separate time windows
Noise	Read the analog sensors only while the pump and GSM radio are idle. Average 32 samples, take the median, RC filter on every input
Timekeeping	DS3231 RTC on I2C (0x68), plus network time when online
Storage	SQLite on the SD card with a read-only root file system; daily CSV export

5. Electrical design

5.1 Raspberry Pi pin map (BCM numbering)

Pi pin	GPIO	Connects to	Notes
1	3.3 V	ADS1115 #1 and #2, DS3231, DHT11, DS18B20 pull-up, rain and LDR modules	About 20 mA in total
2, 4	5 V	(Pi supply input from Buck A, or use micro-USB)	Feeding the 5 V pins bypasses the Pi's input fuse; add an external fuse
3 / 5	GPIO2 / GPIO3	I2C SDA / SCL: both ADS1115s, DS3231	The modules have pull-ups
7	GPIO4	DS18B20 data	4.7 kΩ to 3.3 V

Pi pin	GPIO	Connects to	Notes
8	GPIO14 (TXD)	SIM800L RXD **through a 1 kΩ / 2 kΩ divider**	VIH maximum 3.1 V [4]
10	GPIO15 (RXD)	SIM800L TXD	2.8 V output, read correctly
11	GPIO17	DHT11 data	`dtoverlay=dht11,gpiopin=17`
12	GPIO18	SIM800L RST	Directly, or through an NPN transistor
13	GPIO27	Rain module DO	Module powered at 3.3 V
15	GPIO22	Rain module power switch (NPN to a P-MOSFET)	Powered only while reading
16	GPIO23	MX1508 IN1 (pump)	IN2 tied low
29	GPIO5	LDR module DO	
31	GPIO6	pH board 5 V high-side switch	Never on together with GPIO13
33	GPIO13	TDS board high-side switch	
35	GPIO19	MQ-3 heater MOSFET	About 180 mA heater current
36	GPIO16	ACK / reset button to ground	Internal pull-up
37	GPIO26	Status LED / beacon (through a transistor)	
6, 9, 14 …	GND	Star ground at ADS1115 #1	

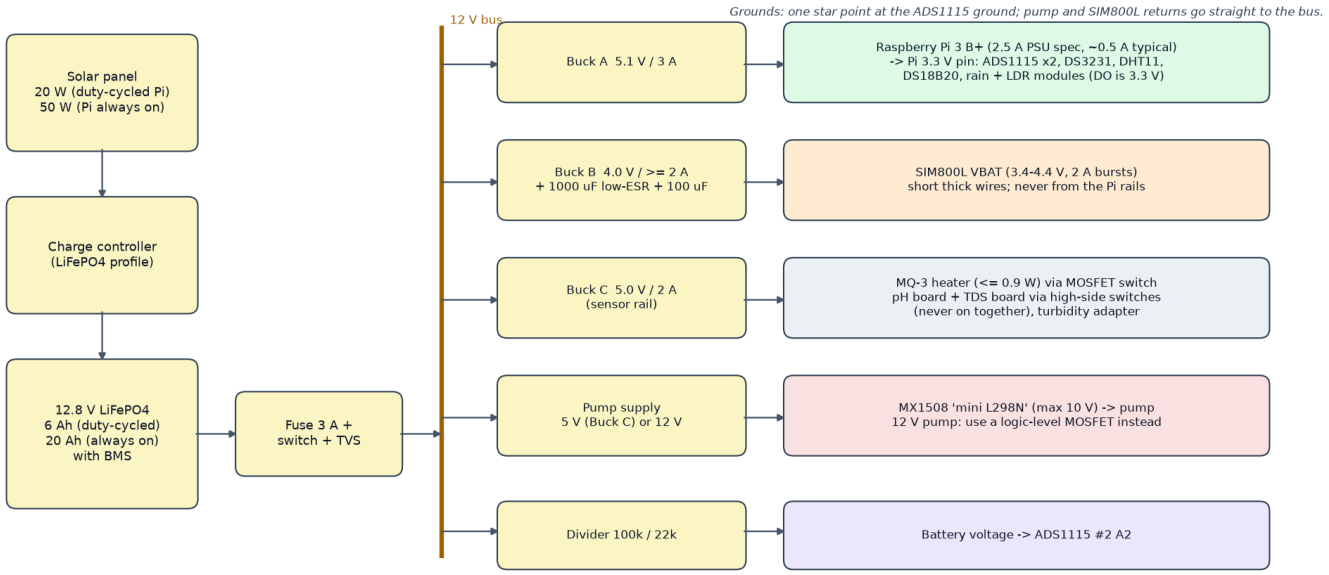
5.2 ADC channels

ADC	Ch	Signal	Raw range	Conditioning	Range at the ADC
#1 0x48	A0	pH Po	0–5 V (centred about 2.5 V)	12 k / 20 k divider, 1 k + 100 nF	0–3.1 V
#1	A1	TDS	0–2.3 V	1 k + 100 nF	0–2.3 V
#1	A2	Turbidity	0–4.5 V	12 k / 20 k, 1 k + 100 nF	0–2.8 V
#1	A3	MQ-3 AO	0–5 V	12 k / 20 k, 1 k + 100 nF	0–3.1 V
#2 0x49	A0	LDR AO	0–3.3 V	100 nF	0–3.3 V
#2	A1	Rain AO	0–3.3 V	100 nF	0–3.3 V
#2	A2	Battery voltage	0–14.6 V	100 k / 22 k divider	0–2.6 V
#2	A3	Spare (DO probe or pump current later)			

To recover the sensor voltage, multiply by 1.6 for a 12 k / 20 k divider, and by 5.55 for the battery divider. The divider's source resistance causes a small gain error; the per-sensor calibration removes it.

5.3 Power architecture and budget

Power architecture (field version; on the bench replace the solar/battery block with a 12 V 3 A adapter)



Load	Power	Always-on: Wh/day	Pi woken 3 min every 15 min: Wh/day
Raspberry Pi 3 B+ (500 mA typical at 5 V) [1]	about 2.5 W	60	12
MQ-3 heater (on continuously in always-on mode; about 1 h/day for tests in timed mode) [5]	about 0.86 W	20.6	0.9
ADCs, RTC, sensor boards (switched)	about 0.15 W	3.6	0.7
SIM800L idle (17 mA at 4 V) plus a few SMS [4]	about 0.07 W	1.7	1.7
Pump (30 s per event)	about 1 W	<0.1	<0.1
Wake-up timer board			about 0.5
Buck converter losses (15 %)		+13	+2.4
Total		**about 100 Wh/day**	**about 19 Wh/day**

Sizing:

Mode	Battery	Panel
Always on	12.8 V 20 Ah LiFePO4 (2 days)	50 W (100 Wh ÷ (4.5 sun-hours × 0.7) ≈ 32 W, plus monsoon margin)
Timed (recommended for the field)	12.8 V 6 Ah (about 3.5 days; the R3's battery)	20 W (the R3's panel)

Timed mode needs a wake-up timer board such as Witty Pi, or a small microcontroller, to switch the Pi on.

On the bench and at the finale: use a 12 V 3 A adapter, or a 5 V 3 A adapter for the Pi plus separate 4 V and 5 V rails.

5.4 Protection, grounding and noise

- **One star ground point** at ADS1115 #1:
  - the analog grounds return there;

    - the pump and SIM800L currents return directly to the supply bus, not through the analog ground.
- **Every ADC input:** 1 kΩ series resistor and 100 nF to ground. On long sensor cables add a BAT54S clamp to 3.3 V and ground.
- **SIM800L:**
    - 1000 µF low-ESR plus 100 nF across VBAT, and a 5.1 V Zener as SIMCom recommends; [4]
    - short wires of at least 0.5 mm^2;
    - an antenna outside the box, away from the analog cables.
- **Pump:** a flyback diode (1N5819) and 100 nF across the motor. Run it only after the analog readings.
- **Main supply:** fuse (3 A), reverse-polarity protection and a TVS diode on the 12 V input.
- **Wiring:** twisted or shielded sensor cables, kept away from the pump and antenna wires.

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6. Sampling mechanism and sensor placement

6.1 Continuous sensing (every cycle)

The pH, TDS, turbidity and DS18B20 probes sit in the **3D-printed circular sensor housing** in `12_SIMPLE_SENSOR_HOUSING`:

- **TDS probe:** goes in the former "EC" port; its probe is smaller, so the bore will be resized.
- **Spare port:** takes the pump intake.

Measure every probe before printing; resizing the bores is a one-minute rebuild.

6.2 Event sample and MQ-3 headspace test

1. The alert logic confirms an event (section 7.4).
2. The pump fills the 250 mL bottle to about **150 mL**, leaving about 100 mL of air:
    - time-based fill, calibrated on the bench (about 6 s at 100 L/h);
    - an optional float switch stops overfill.
3. The bottle is sealed. Its lid carries the MQ-3, which has been preheated because the heater is already on.
4. Wait **5 minutes** for vapours to equilibrate, then read Rs/R0 every 10 s for 2 minutes. Record the minimum.
5. Log the result, send the SMS, and label the bottle with the time for the lab. One bottle per event; a replaced bottle re-arms sampling.

Caveats:

- The MQ-3 heater warms the headspace slightly. For a lab-grade BOD sample, use a second, cooled bottle.
- The tubing holds about 20 mL from the previous cycle. Run the pump for 2 s to waste first, if the tubing allows it.

6.3 Avoiding pH and TDS interference

The TDS board drives an AC signal into the water, while the pH probe is a high-impedance electrode. Powered together they share a current path through the water and their common ground. DFRobot documents this and sells isolators as the fix. [17]

The design instead switches each board's supply with a P-MOSFET high-side switch (AO3401 plus an NPN driver):

1. **TDS window:** TDS on for 5 s, then read.

2. **TDS off.**
3. **pH window:** pH on for 30 s to settle, then read.

Bench test: compare the pH reading with the TDS board switched on and off. If the difference stays above 0.05 pH, add DFRobot DFR0504 isolators at ₹2,219 each.

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7. Software design

7.1 Operating system configuration

Raspberry Pi OS Lite (Bookworm). In `/boot/firmware/config.txt`:

```
dtparam=i2c_arm=on
dtoverlay=w1-gpio,gpiopin=4
dtoverlay=dht11,gpiopin=17
enable_uart=1
dtoverlay=disable-bt
dtoverlay=i2c-rtc,ds3231
dtparam=watchdog=on
```

`disable-bt` gives the reliable PL011 UART on GPIO14/15. Then:

- disable `hciuart`;
- in `raspi-config`, turn the serial login shell off and the serial hardware on;
- in `raspi-config`, enable the overlay file system (read-only root) before deployment. [2][12]

7.2 Services (Python 3, systemd, `Restart=always`)

Service	Job	Libraries
`acquire`	Runs the measurement cycle (section 8), filters readings, applies calibration	adafruit-circuitpython-ads1x15, gpiozero, sysfs (1-Wire, DHT11)
`decide`	Hard limits, hourly baselines, 2-of-3 confirmation, events	Plain Python (the R3 algorithm, ported)
`store`	SQLite tables `readings`, `events`, `samples`; daily CSV	sqlite3
`comms`	SIM800L SMS and HTTP over GPRS; MQTT over WiFi; retry queue	pyserial, paho-mqtt
`sampler`	Pump control, one bottle per event, MQ-3 headspace test	gpiozero
`health`	Watchdog, disk and battery checks, daily status SMS	

Data flow: sensors → `acquire` → SQLite → `decide` → events → `sampler` / `comms`.

- **Dashboard:** MQTT topic `aquasentinel/<site>/readings` → Node-RED or Grafana on a laptop for the finale, a cloud broker later.
- **SMS commands:** `STATUS`, `SAMPLE`, `RESET` from approved numbers only.

7.3 SIM800L commands

- **Start-up:** `AT`, `AT+CPIN?`, `AT+CSQ` (signal 10 or more is usable), `AT+CREG?`.
- **SMS:** `AT+CMGF=1`, then `AT+CMGS="+91…"`, the text and Ctrl-Z.
- **GPRS:** `AT+SAPBR=3,1,"APN","<operator APN>"`, `AT+HTTPINIT`, `AT+HTTPPARA="URL",…`, then `AT+HTTPACTION=1`.
- **Network time:** `AT+CLTS=1`.
- Keep phone numbers and server keys in a local configuration file, never in the repository.

7.4 Alert logic

Parameter	Watch	Alarm	Basis
pH	outside 6.5–8.5	outside 6.0–9.0	CPCB water-quality criteria; the R3 limits [18]
TDS	+25 % over the hourly baseline	above 900 ppm (near the sensor's ceiling) or +50 %	Sensor range [6]; the CPCB irrigation limit (EC 2,250 µS/cm) is beyond this sensor
Turbidity	+25 % over baseline (estimated NTU)	+50 %	Relative, because readings are uncalibrated
Water temperature	+2 °C over the hourly baseline	+3 °C	Thermal discharge
MQ-3 headspace	Rs/R0 below 0.8	below 0.6	Relative, from clean-air R0
Rain	—	—	Context only: raises the turbidity thresholds and tags events "runoff-likely"

- **Event:** 2 of the last 3 cycles abnormal on any parameter, or any alarm confirmed on 2 consecutive cycles.
- **Actions:**
    1. Log it.
    2. Turn the beacon red.
    3. Send an SMS to the list.
    4. Send an HTTP or MQTT message.
    5. Take one sample, with the MQ-3 test.
- **Lockout:** no second sample until the event clears or the bottle is replaced.
- **Faults:** a dead or disconnected sensor raises **SENSOR FAULT**, never "normal". This is the R3 audit's fail-safe fix.

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8. Measurement cycle

Every 15 minutes in the field; every 15 s in demo mode, labelled as such.

Time	Step
0 s	Wake: read the DS3231 time and battery voltage; DS18B20 conversion (750 ms)
1 s	DHT11 read; rain module on, read its outputs, off; LDR read
2 s	Turbidity: 32 samples, median
3 s	TDS window: TDS on, 5 s settle, 32 samples, off
9 s	pH window: pH on, 30 s settle, 32 samples, off (with temperature compensation)
40 s	Compute, store, decide
41 s	Communicate: GSM or MQTT; pump only if an event was confirmed
about 60 s	Sleep (timed mode: the Pi shuts down until the next wake)

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9. Calibration

Sensor	Procedure	How often
pH	1. Offset: short the BNC, trim Po to 2.50 V. 2. Two-point calibration in pH 6.86/7.00 and 4.01 (add 9.18 for alkaline water). 3. Record slope and offset in the configuration	Before each demo day; weekly in the field

Sensor	Procedure	How often
TDS	Measure a reference solution: 707 ppm (1,413 µS/cm KCl) or a meter-checked sample. Adjust the K factor. Temperature-compensate to 25 °C	Every 2 weeks
Turbidity	Measure clear filtered water (about 0 NTU) and two dilutions of a cloudy suspension, cross-checked against a lab turbidimeter if possible. Fit a curve	Monthly; clean the optics weekly
MQ-3	48 h burn-in. R0 in clean air at known temperature and humidity. Check the response with a drop of hand sanitiser in a sealed jar	Re-baseline weekly
DS18B20	Ice water (0 °C) and room temperature against a reference thermometer	Once, then yearly
DHT11	Compare against a reference hygrometer	Once
Pump	Time how long it takes to fill 150 mL and 250 mL on the actual tubing and lift	After any tubing change

10. Enclosure, waterproofing and environmental protection

Electronics box: IP65/IP67 ABS or PC enclosure, about 250 × 200 × 120 mm.

- **Inside:**
  - the Pi and an ADC carrier board (perfboard with screw terminals);
  - the three buck converters;
  - the SIM800L, close to an SMA bulkhead with the antenna outside;
  - a fuse block.
- **Glands:** M16/PG7 cable glands, with drip loops on every cable.
- **Moisture:** a breathable vent (Gore-type) and a desiccant pack.
- **Heat:** mount it in shade.

Outside the box:

Item	Placement
DHT11	Louvered radiation shield
Rain plate	On top, tilted about 30° to shed water
LDR	Facing the sky behind a clear window, not shaded by the panel
Battery	Separate box, never in the electronics box

Water side:

- The printed sensor housing holds the probes. Probe cables reach the box through glands.
- The turbidity sensor's top must stay above the waterline (the housing's fill grooves).
- The pump intake has a strainer.

Corrosion:

- stainless or nylon fasteners;
- conformal coating on the boards, never on sensing parts;
- the rain comb powered only while reading;
- the pH probe never allowed to dry.

Field risks: floods (a lift-out procedure), vandalism (a lockable box, low visibility), lightning (TVS diodes, short cables).

11. Maintenance plan

How often	Task
Daily (pilot)	Check the status SMS: battery, last reading, signal strength
Weekly	Clean the pH, TDS and turbidity probes (soft brush, rinse). pH one-point check at 7.00. Clean the turbidity optics. Remove the sample bottle. Check the desiccant
Every 2 weeks	Full pH calibration; TDS check; refill the pH storage solution
Monthly	Turbidity check; MQ-3 re-baseline; inspect glands and seals; back up the SD card
Every 6–12 months	Replace the pH probe (sooner in dirty water) and the rain comb if corroded

12. Bill of materials: what to add

Existing hardware: all 12 items in section 1, which the team already has.

Essential for the bench prototype (Robu.in, prices including GST, checked 28 Oct 2026; "est." means not checked):

Item	Qty	Price	Link
ADS1115 16-bit 4-channel ADC module	2	₹129 each	Robu
DS3231 RTC module	1	₹125	Robu
5 V 5 A buck converter (Pi supply, Buck A)	1	₹199	Robu
LM2596 buck converter with display (SIM800L 4.0 V, Buck B)	1	₹129	Robu
LM2596S buck converter (5 V sensor rail, Buck C)	1	₹49	Robu
4-channel level converter (optional; a resistor divider also works)	1	₹39	Robu
Resistor kit (12 k, 20 k, 1 k, 2 k, 4.7 k, 100 k, 22 k)	1	₹489	Robu
1000 µF 16 V low-ESR capacitor, 100 nF ceramic capacitors, 1N5819 diodes, 5.1 V Zener	set	about ₹150 (est.)	Any local shop
AO3401 P-MOSFET × 3, 2N3904 × 4, IRLZ44N × 1 (power switches)	set	about ₹120 (est.)	Any local shop
Perfboard, screw terminals, female headers	set	about ₹200 (est.)	Any local shop
12 V 3 A adapter and DC jack (bench)	1	about ₹350 (est.)	Any local shop
32 GB high-endurance microSD	1	about ₹600 (est.)	Any local shop
pH buffer set (4.01 / 6.86 / 9.18)	1	about ₹250 (est.)	Any local shop
Bench total		**about ₹3,000**	

For the field pilot only (prices from the team's 23 Oct cost research unless marked est.):

- LiFePO4 12.8 V 6 Ah battery: ₹2,200; or 20 Ah for always-on, about ₹6,000 (est.).
- 20 W solar panel: ₹1,299.
- Charge controller: about ₹600 (est.).
- IP67 enclosure: ₹1,500.
- Cable glands: ₹70 each.
- Breathable vent: about ₹150 (est.).
- Wake-up timer board: about ₹3,000 (est.).

Scientifically justified optional additions:

Item	Gap it closes	Price
Raspberry Pi Camera Module 3	Floating plastic and solid waste; visual evidence for events	About ₹3,280 including GST (Evelta, NoIR version listed)

Item	Gap it closes	Price
Dissolved oxygen sensor	Sewage and organic pollution	Optical RS485 type ₹19,040 ex-GST (Evelta); a cheaper galvanic type needs membrane upkeep
4G LTE Cat-1 module (SIMCom A7670C)	Removes the 2G risk	About ₹1,900 (R3 cost research)
DFRobot DFR0504 isolators	Only if switching the supplies doesn't remove the pH/TDS interference	₹2,219 each (Robu)

13. Implementation and test roadmap

The finale exhibition is on 19 Nov.

Phase	Dates (2026)	Work	Done when
0. Prepare	28–31 Oct	Answer section 15. Order the section 12 parts. Flash Raspberry Pi OS and apply the section 7.1 configuration. **Start the MQ-3's 48 h burn-in on the day the parts arrive**	`i2cdetect` shows 0x48, 0x49 and 0x68; the DS18B20 appears under `/sys/bus/w1`
1. One sensor at a time	1–6 Nov	DS18B20 → DHT11 → ADS1115 + TDS → pH (calibrate) → turbidity → MQ-3 → rain and LDR → pump (fill time) → SIM800L SMS (signal strength)	Each sensor reads plausibly in tap water, salt water, vinegar and muddy water; an SMS is received
2. Integrate	7–12 Nov	Switched pH and TDS supplies (interference test). Measurement cycle, SQLite, alert logic, sampler, MQTT dashboard. Print the sensor housing with the measured bores. 24 h unattended run on the bench	No reset during GSM bursts; the pH change with TDS on/off is under 0.05; 24 h log without gaps
3. Demo hardening	13–17 Nov	Demo mode (15 s cycle), read-only root file system, watchdog, spare probe and bottle, demo script with salt, vinegar, muddy water and sanitiser. Three full rehearsals	Three clean runs back to back; recovers after the power is pulled
4. Field pilot	After 20 Nov	Enclosure, solar, timed waking, a 2-week supervised trial at a drain or canal with lab samples for comparison	Over 90 % of readings received; events match the lab results
5. Upgrade path	2027	4G LTE, dissolved oxygen, industrial probes: converge on the R3 field design	—

14. Risk register

Risk	Likelihood	Impact	Mitigation
SIM800L brown-outs reset the module or the Pi	High	High	Dedicated 4.0 V rail with 2 A capacity, 1000 µF, short wires (section 5.4)
pH reading disturbed by the TDS board	High	High	Switched supplies and time windows; isolators if needed
No 2G at the site	Medium	High	Check with `AT+CSQ`; Airtel or Vi SIM; store and forward over WiFi; 4G upgrade
SD-card corruption after a power cut	Medium	High	Read-only root, data on a separate partition, a safe-shutdown script
MQ-3 false alarms from humidity or temperature	High	Medium	Sealed headspace protocol, relative threshold, DHT11 context, "indicator" wording
Turbidity affected by light or bubbles	Medium	Medium	Measure inside the housing, out of the sun; median filter
pH probe dries out or drifts	Medium	High	Wet storage, calibration schedule, spare probe
Pump runs dry or sucks debris	Medium	Medium	Intake strainer; run only when submerged; time limit
Water in the electronics box	Medium	High	IP67 box, glands, drip loops, vent, desiccant
Field power shortfall with the Pi always on	High	High	Timed waking (19 Wh/day)
Overclaiming to the jury	Medium	High	"Indicators, not identification"; "prototype"; the R3 is the field design

15. Missing information (please confirm)

#	Question	Default used here
1	Exact pH board and probe (photo of the board)?	PH-4502C with E-201 probe (A1)
2	Exact turbidity and TDS boards (photo, markings)?	SEN0189 and SEN0244 types (A2, A3)
3	Pump voltage and flow (label on the pump)?	3–6 V, about 100 L/h (A4)
4	SIM800L board: red mini module or blue "V2 5V" board? Which SIM operator?	Red mini module; Airtel or Vi (A5)
5	MQ-3 module load resistor (RL) value?	Measure it before calibrating
6	Finale demo powered from mains? Field site and dates?	Mains at the finale; supervised pilot afterwards (A7)
7	Who receives alerts, and which dashboard (laptop or cloud)?	Team phones; Node-RED on a laptop
8	Lab partner for the samples?	To arrange before the pilot

16. Next actions (this week)

1. **Team:** answer section 15. Photos of the pH, TDS, turbidity and SIM800L boards are enough.
2. **Team:** order the section 12 essentials (about ₹3,000) from Robu, which is in Pune.
3. **Team:** flash Raspberry Pi OS Lite and apply section 7.1. Start the MQ-3 burn-in as soon as it's powered.
4. **Team:** measure the probe diameters with calipers, then resize the printed sensor housing.
5. **Team:** write the Python services in section 7, the wiring sheet for the carrier board, and the demo script.

17. References

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